

DP Computer Science: A 3.3.1 Data Language Types

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- **Outline** the difference between Data Definition Language (DDL) and Data Manipulation Language (DML)
 - **Identify** SQL statements that are classified as DDL
 - **Identify** SQL statements that are classified as DML
 - **Use** `CREATE TABLE` in DB Browser for SQLite to define a simple table
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Key Vocabulary

Term	Definition
SQL	Structured Query Language; the standard language for interacting with relational databases
DDL (Data Definition Language)	SQL commands used to define and manage the structure of a database (schema)
DML (Data Manipulation Language)	SQL commands used to manipulate the actual data stored in a database
CREATE TABLE	A DDL command that creates a new table in the database
ALTER TABLE	A DDL command that modifies the structure of an existing table

Term	Definition
DROP TABLE	A DDL command that deletes a table from the database
SELECT	A DML command that retrieves data from a database
INSERT INTO	A DML command that adds new records to a table
UPDATE	A DML command that modifies existing records in a table
DELETE FROM	A DML command that removes records from a table
Schema	The structure of a database, including tables, columns, and relationships

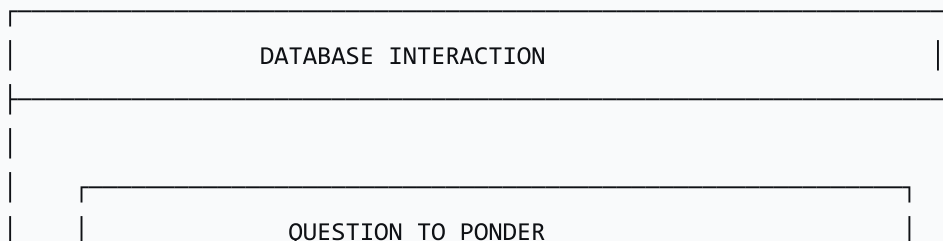
IB Syllabus Link

A3 Databases | A3.3 Database programming | A3.3.1 Outline the differences between data language types within SQL.

2. Why Do We Need Different Language Types?

The Two Aspects of Database Interaction

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"How do we both define the structure of a database and then work with the data inside it?"

Two fundamental aspects:

STRUCTURE
(The "Container")

- Tables
- Columns
- Data types
- Constraints

DATA
(The "Contents")

- Records (rows)
- Values
- Actual information

Different types of SQL commands are needed for each:

DDL
Defines structure

DML
Manipulates data

3. Data Definition Language (DDL)

What is DDL?

DDL (Data Definition Language) is the set of SQL commands used to define and manage the structure of the database. It deals with the schema—the tables, columns, data types, and constraints.

Key DDL Commands

Command	Purpose	Example
CREATE TABLE	Creates a new table	<code>CREATE TABLE Students (...);</code>

Command	Purpose	Example
ALTER TABLE	Modifies an existing table structure	ALTER TABLE Students ADD COLUMN Email TEXT;
DROP TABLE	Deletes a table	DROP TABLE Students;
CREATE INDEX	Creates an index for faster queries	CREATE INDEX idx_name ON Students(LastName);
DROP INDEX	Deletes an index	DROP INDEX idx_name;

DDL Focus: What DDL Does

text

DDL COMMANDS

```
CREATE TABLE Students
(
    StudentID INTEGER PRIMARY KEY,
    FirstName TEXT NOT NULL,
    LastName TEXT NOT NULL,
    DateOfBirth DATE,
    GradeLevel INTEGER
);
```

- ✅ Defines the table structure
- ✅ Specifies column names and data types
- ✅ Sets constraints (PRIMARY KEY, NOT NULL)
- ❌ Does NOT add or modify data

CREATE TABLE Example

Component	Description	Example
Table Name	The name of the new table	Students
Columns	Each column's name, data type, and constraints	StudentID INTEGER PRIMARY KEY
PRIMARY KEY	Uniquely identifies each record	StudentID
NOT NULL	Ensures the column cannot be empty	FirstName TEXT NOT NULL

Key Points about DDL

Point	Description
Schema Changes	DDL changes the structure, not the data
Auto-Commit	DDL commands are often auto-committed (cannot be rolled back in some databases)
Metadata	DDL creates and modifies metadata about the database
One-Time Setup	DDL is typically used during database design and setup

4. Data Manipulation Language (DML)

What is DML?

DML (Data Manipulation Language) is the set of SQL commands used to manipulate the actual data stored within the database structures. It deals with the content—the records in the tables.

Key DML Commands

Command	Purpose	Example
SELECT	Retrieves data from a table	SELECT * FROM Students;

Command	Purpose	Example
INSERT INTO	Adds new records to a table	<code>INSERT INTO Students VALUES (...);</code>
UPDATE	Modifies existing records	<code>UPDATE Students SET GradeLevel = 11 WHERE StudentID = 1;</code>
DELETE FROM	Removes records from a table	<code>DELETE FROM Students WHERE StudentID = 5;</code>

DML Focus: What DML Does

text

DML COMMANDS

```
INSERT INTO Students
(StudentID, FirstName, LastName)
VALUES
(1, 'Alice', 'Smith'),
(2, 'Bob', 'Jones'),
(3, 'Charlie', 'Brown');
```

```
SELECT * FROM Students
WHERE GradeLevel = 11;
```

- ✅ Adds, retrieves, modifies, and deletes data
- ✅ Works with the actual content of the tables
- ❌ Does NOT change the table structure

DML Examples

SELECT — Retrieving Data:

```
sql

-- Select all students
SELECT * FROM Students;

-- Select specific columns
SELECT FirstName, LastName FROM Students;

-- Select with a condition
SELECT * FROM Students WHERE GradeLevel = 11;
```

INSERT INTO — Adding Data:

```
sql

-- Insert a new student
INSERT INTO Students (StudentID, FirstName, LastName, GradeLevel)
VALUES (4, 'Diana', 'Williams', 10);
```

UPDATE — Modifying Data:

```
sql

-- Update a student's grade level
UPDATE Students
SET GradeLevel = 12
WHERE StudentID = 4;
```

DELETE FROM — Removing Data:

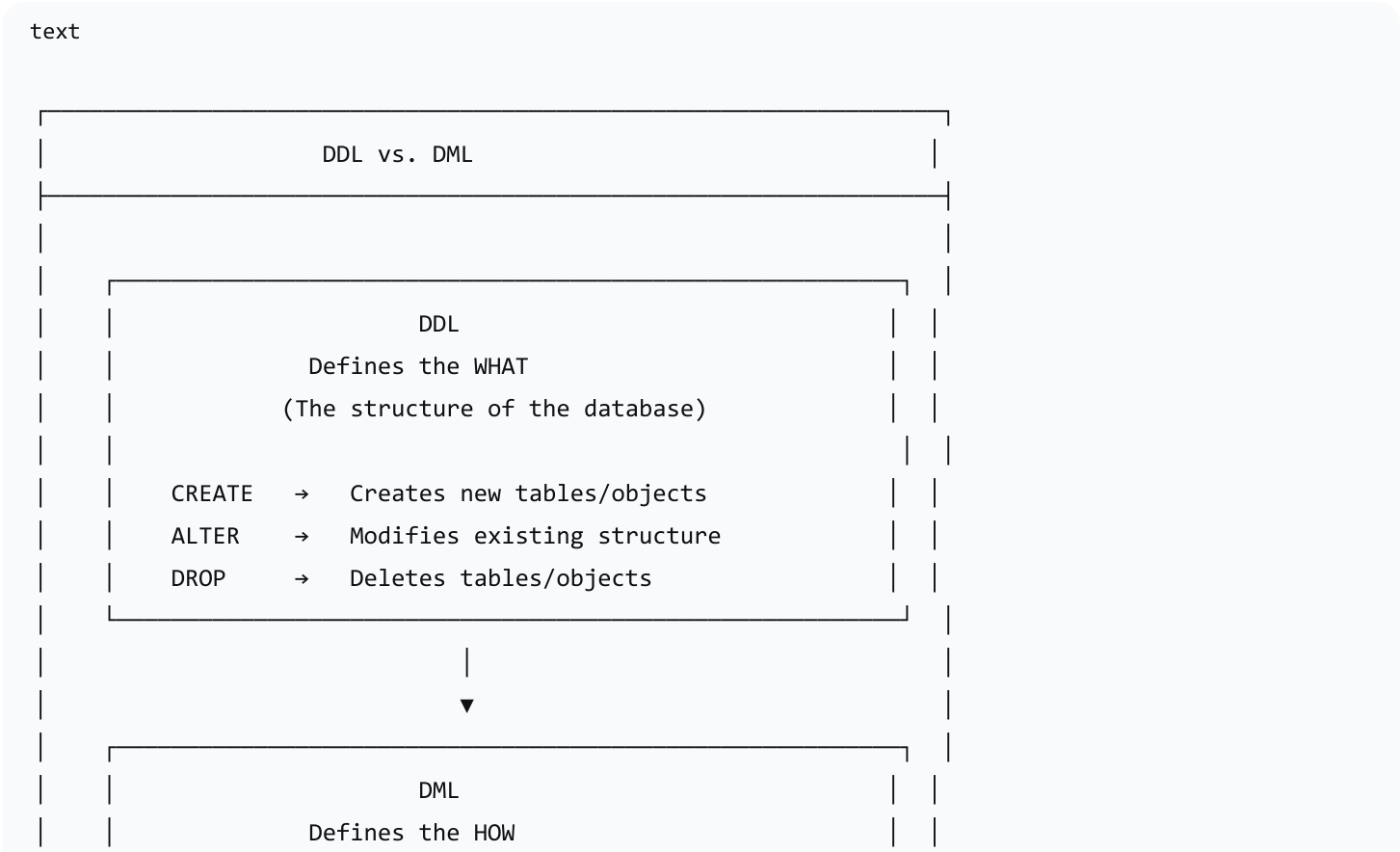
```
sql

-- Delete a student
DELETE FROM Students
WHERE StudentID = 4;
```

5. DDL vs. DML: Key Differences

Aspect	DDL	DML
Purpose	Defines and manages database structure	Manages the actual data
Commands	CREATE, ALTER, DROP	SELECT, INSERT, UPDATE, DELETE
Works On	Schema (tables, columns, constraints)	Data (records, values)
Changes	Changes the structure	Changes the content
Auto-Commit	Usually auto-committed	Can be rolled back (transactions)
Frequency	Less frequent (during design/setup)	More frequent (ongoing operations)
Analogy	Building the house	Furnishing and living in the house
Examples	CREATE TABLE , ALTER TABLE	SELECT , INSERT INTO , UPDATE , DELETE

Visual Summary



(The manipulation of data)

SELECT	→	Retrieves data
INSERT	→	Adds new data
UPDATE	→	Modifies existing data
DELETE	→	Removes data

6. Activity: Creating a Table in DB Browser for SQLite

Instructions

1. Open DB Browser for SQLite
2. Create a new database or open an existing one
3. Use the "Execute SQL" tab or the "Create Table" tab
4. Write a `CREATE TABLE` statement for a table of your choice

Example: Books Table

sql

```
CREATE TABLE Books (  
    BookID INTEGER PRIMARY KEY,  
    Title TEXT NOT NULL,  
    Author TEXT NOT NULL,  
    YearPublished INTEGER,  
    Genre TEXT  
);
```

Example: Students Table

sql

```
CREATE TABLE Students (  
    StudentID INTEGER PRIMARY KEY,  
    FirstName TEXT NOT NULL,  
    LastName TEXT NOT NULL,  
    DateOfBirth DATE,  
    GradeLevel INTEGER,  
    IsEnrolled BOOLEAN  
);
```

Example: Movies Table

```
sql  
  
CREATE TABLE Movies (  
    MovieID INTEGER PRIMARY KEY,  
    Title TEXT NOT NULL,  
    Director TEXT,  
    ReleaseYear INTEGER,  
    Rating REAL  
);
```

Student Task

Choose one of the following tables to create in DB Browser for SQLite:

Option	Table Name	Columns
Option 1	Books	BookID, Title, Author, YearPublished, Genre
Option 2	Students	StudentID, FirstName, LastName, DateOfBirth, GradeLevel
Option 3	Movies	MovieID, Title, Director, ReleaseYear, Rating

Requirements:

- At least **three columns**
- Specify appropriate **data types** (TEXT, INTEGER, REAL, DATE, BOOLEAN)
- Include a **PRIMARY KEY** constraint

- Use NOT NULL where appropriate

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	5 min	Welcome; introduce database programming with SQL; state learning objectives
Introducing SQL	10 min	Review database purpose; introduce SQL; guiding question on structure vs. data
DDL Deep Dive	10 min	CREATE TABLE, ALTER TABLE, DROP TABLE; constraints; data types
DML Deep Dive	10 min	SELECT, INSERT INTO, UPDATE, DELETE FROM; filtering and conditions
Activity	10 min	Students create a simple table in DB Browser for SQLite
Review	10 min	Compare DDL and DML; classify SQL statements; key differences
Conclusion	5 min	Summarise; assign follow-up task

Activity: Classify the SQL Statement

SQL Statement	DDL or DML?
<code>CREATE TABLE Employees (...);</code>	DDL
<code>SELECT * FROM Employees;</code>	DML

SQL Statement	DDL or DML?
<code>INSERT INTO Employees VALUES (...);</code>	DML
<code>ALTER TABLE Employees ADD COLUMN Email TEXT;</code>	DDL
<code>UPDATE Employees SET Salary = 50000 WHERE ID = 1;</code>	DML
<code>DROP TABLE Employees;</code>	DDL
<code>DELETE FROM Employees WHERE ID = 5;</code>	DML

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
Activity	Assess ability to create a table using CREATE TABLE
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the difference between DDL and DML?
2. Give one example of a DDL command and one example of a DML command.
3. What does the CREATE TABLE statement do?
4. Which SQL command would you use to retrieve data from a table?

9. Differentiation

Level	Strategy
Support	Provide pre-written CREATE TABLE examples; use visual aids and step-by-step instructions

Level	Strategy
Challenge	Encourage research on additional DDL/DML commands or explore constraints in more detail

10. Resources

- Presentation (Slide Deck)
- eBook (A3.3.1 with key questions and quiz)
- DB Browser for SQLite (software for hands-on activity)
- Student Activity (instructions for creating tables)

11. Summary: Key Takeaways

Concept	Key Point
DDL	Defines the database structure (schema)—CREATE, ALTER, DROP
DML	Manipulates the data—SELECT, INSERT, UPDATE, DELETE
DDL Command	Changes the structure, not the data
DML Command	Changes the data, not the structure
CREATE TABLE	DDL command for creating new tables
SELECT	DML command for retrieving data

"DDL defines the **what** (the structure of the database), while DML defines the **how** (the manipulation of the data within that structure)."